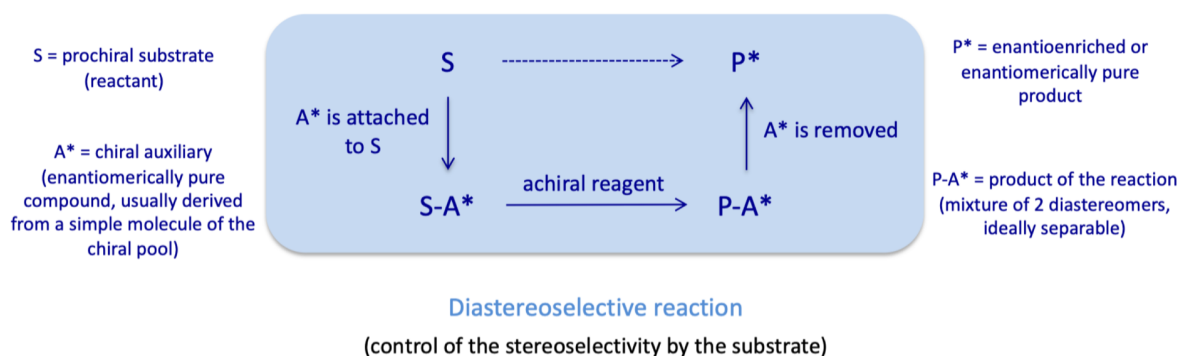


Chap 4: Asymmetric synthesis: An introduction

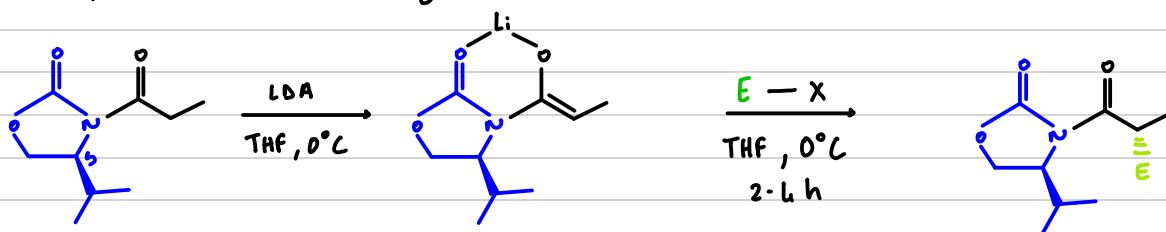
In asymmetric synthesis, stereoselectivity is mainly influenced by the chiral auxiliary, the chiral reagent or the chiral catalyst.

1. The chiral auxiliary strategy



Advantages: the control of the diastereoselectivity is generally good - the obtained diastereomers can often be separated.

Disadvantages: multi-sep reactⁿ · stoichiometric reaction - need to have access to both enantiomers of the chiral auxiliary.



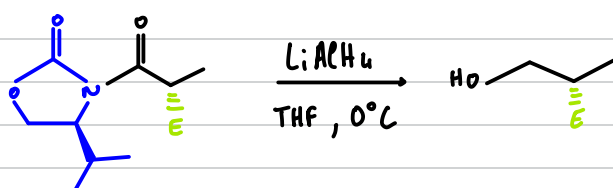
Evans chiral auxiliary

Cis enolate
(Z enolate)
on obtient
uniquement l'énolate cis
c'est le model de Halland

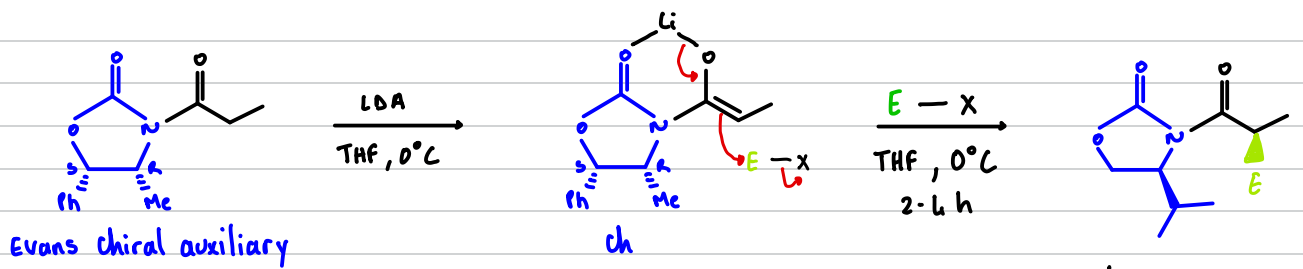
Major diastereomer

E = PhCH ₂	d.r. : 99:1	92%
CH ₂ =CH=CH ₂	d.r. : 98:2	74%
Et	d.r. : 94:6	36%

isolated yield for the major diastereomer



- ici on a obtenu un alcool, mais si on voulais un aldéhyde il fallait utiliser du DIBAL, pour avoir un ester on aurait utilisé un alcoolate.



Major diastereomer

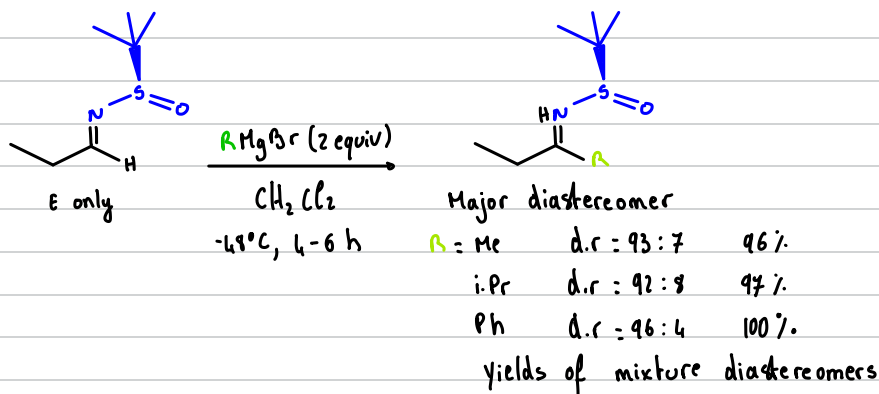
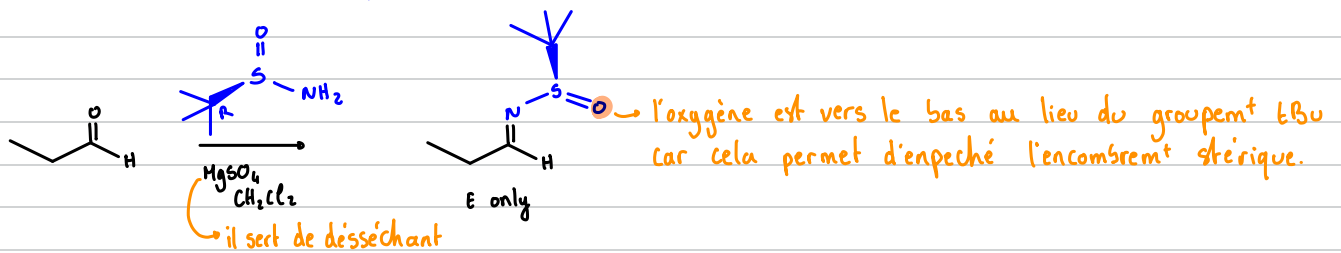
E = PhCH₂

CH₂=CH=CH₂

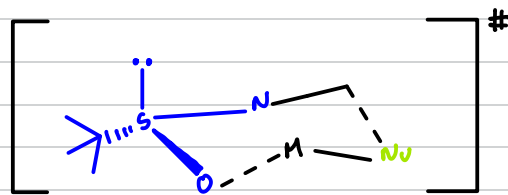
Et

isolated yield for the major diastereomer

Ellman chiral auxiliary



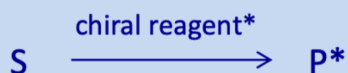
Zimmerman-Traxler
Chelated transit^o-state



Addition on the Si face

2. The chiral reagent strategy

S = prochiral substrate
(reactant)



P* = enantioenriched
product

chiral reagent* =
enantiopure reagent,
used in stoichiometric amount

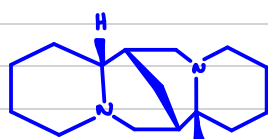
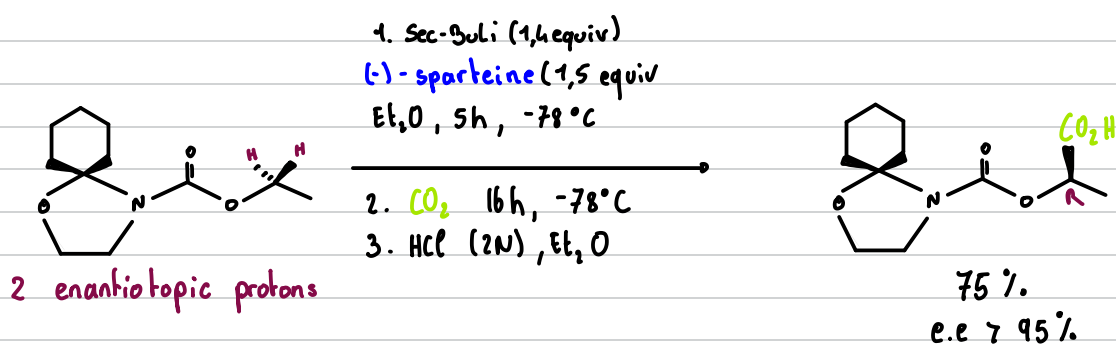
Enantioselective reaction

(control of the stereoselectivity by the reagent)

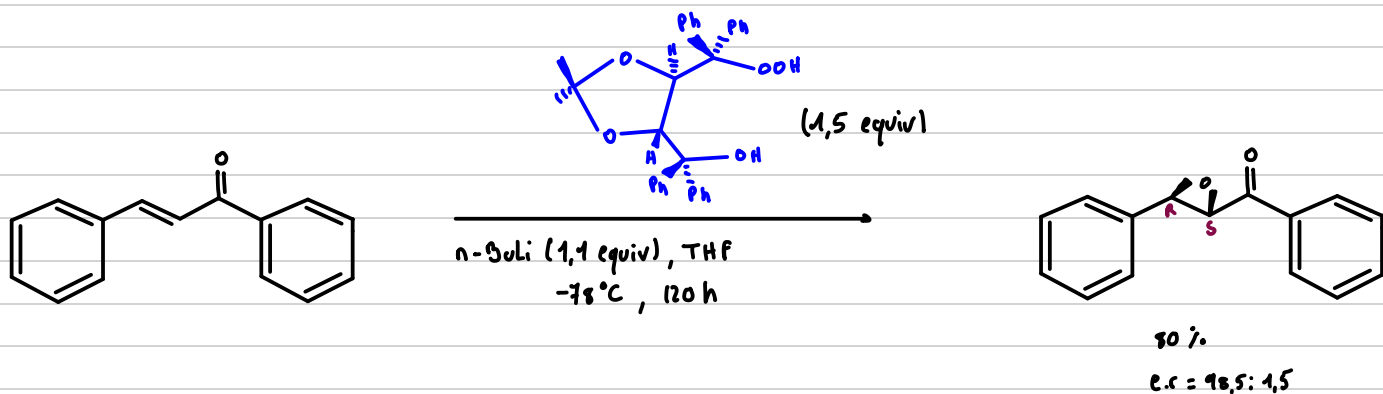
Advantages: One step reaction

↳ greener approach than the chiral auxiliary approach

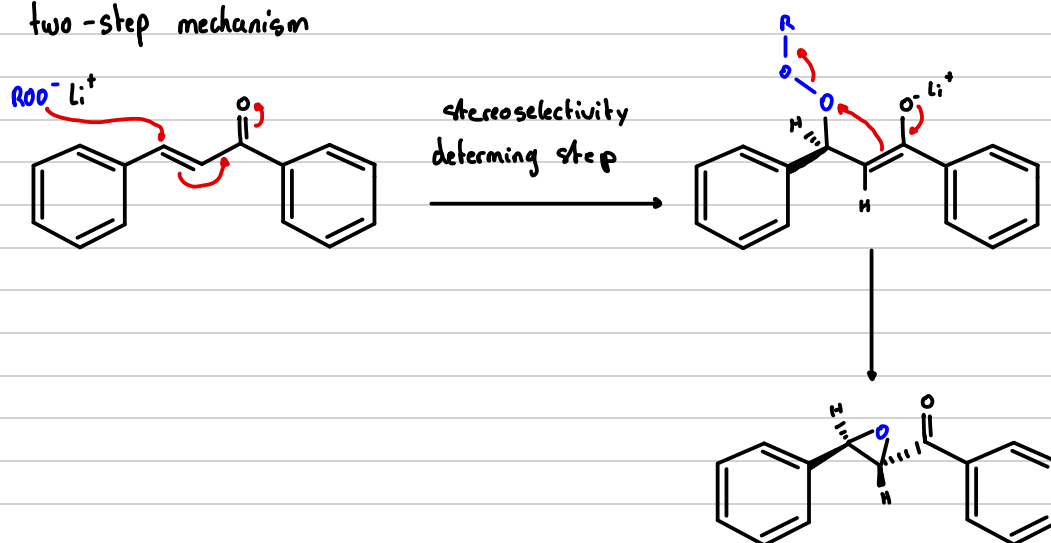
Disadvantages: stoichiometric amount of reagent required - in most cases the reagent need to be prepared - product obtained as mixture of enantiomers which could not be separated easily.



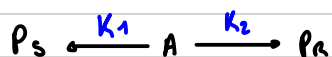
(-)-sparteine



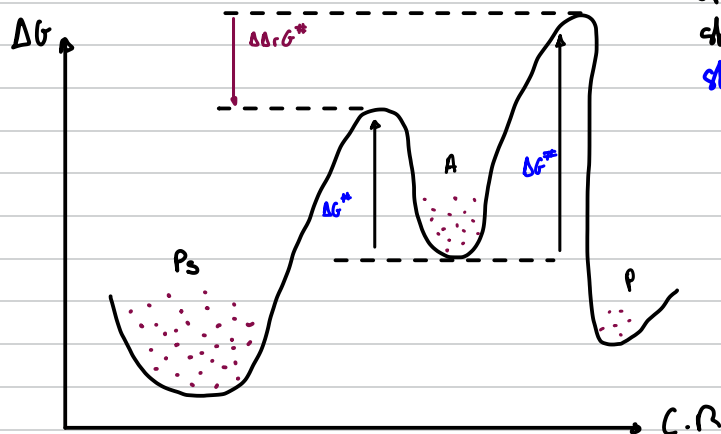
two-step mechanism



Enantioselective reactions are under kinetic control



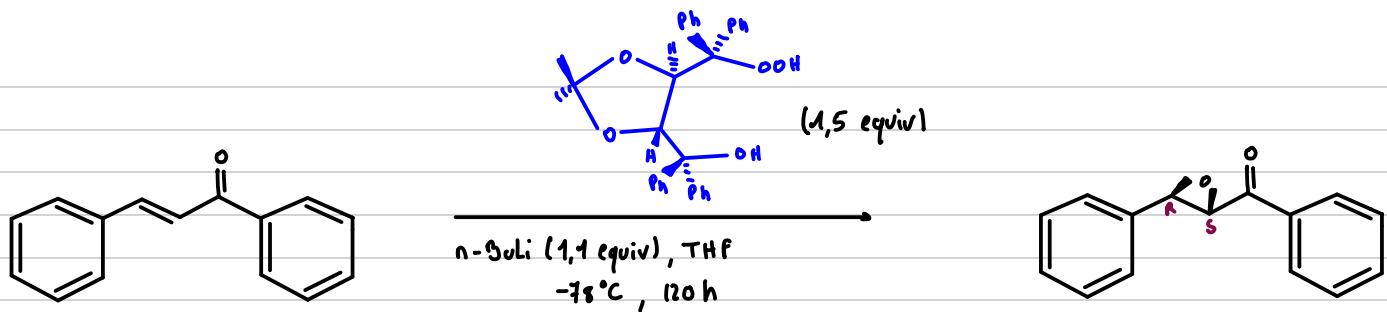
One step mechanism



the use of chiral reagent favors the format of one enantiomer over the other one through interact at the transition state (the T_s must be necessarily diastereoisomeric $\Delta\Delta G^\ddagger \neq 0$)

$$\frac{[P_S]}{[P_R]} = \frac{k_1}{k_2} = e^{-\frac{\Delta\Delta G^\ddagger}{RT}}$$

the temperature of the reaction has an influence on the enantiomeric ratio.

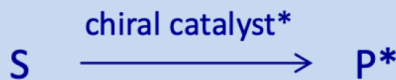


0°C 6h 96% e.r. = 90:10
 -30°C 24h 92% e.r. = 95:5
 -78°C 120h 80% e.r. = 98,5:1,5

Quand on effectue des réact° énantiométriques on ne chauffe pas.

3. The chiral catalyst strategy

S = prochiral substrate (reactant)



P* = enantioenriched product

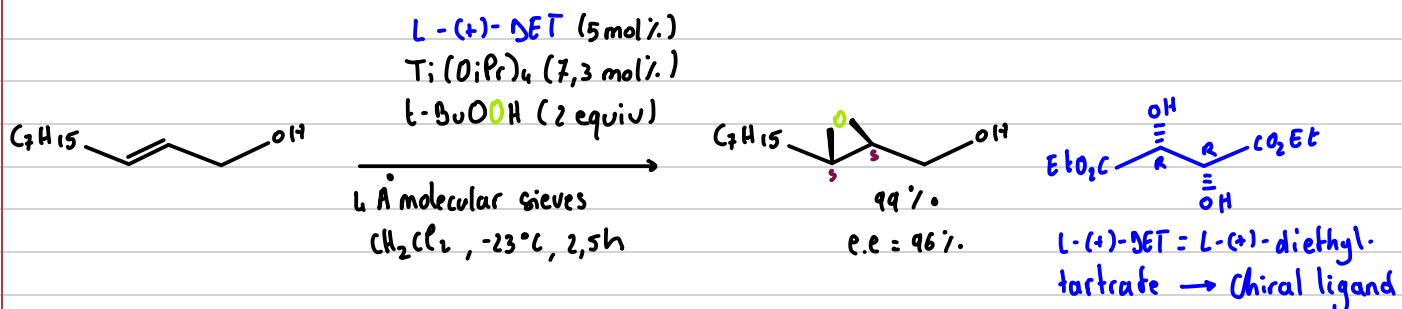
chiral catalyst* =
 enantiopure reagent or ligand,
 used in catalytic amount (at the end of the reaction
 the catalyst should return to its initial state)

Enantioselective reaction

(control of the stereoselectivity by the catalyst)

Advantages: Very elegant method - one step reaction - use of chiral reagent in very low loading.
 ↳ greenest strategy (catalysis is one of the 12 principles of green chemistry)

Disadvantages: highly specific catalyst - identification of the best catalyst is time-consuming - catalysts are often expensive - product obtained as mixture of enantiomers which could not be separated easily.



transition metal-catalysed pathway (chapter 2).

Chapter 2: Enantioselective catalysis: transition

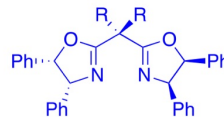
Metal-catalysed pathway

1. Bis(oxazoline) chiral ligands (Box)

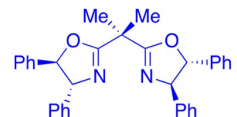
1.1. Structure of bis(oxazoline) chiral ligands



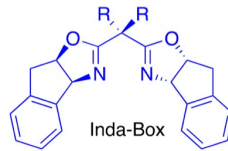
R = H, R' = Ph, Bn, t-Bu, i-Pr (Masamune 1990)
 R = Me, R' = t-Bu, i-Pr (Evans 1991)
 R = Me, R' = Ph (Corey 1991)



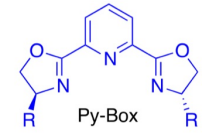
R = H, Me



C₂-symmetric ligands

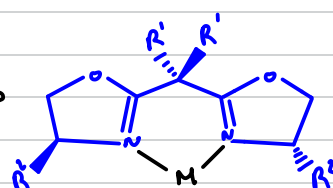
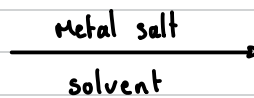
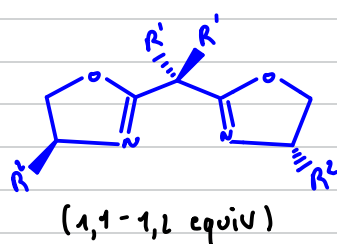
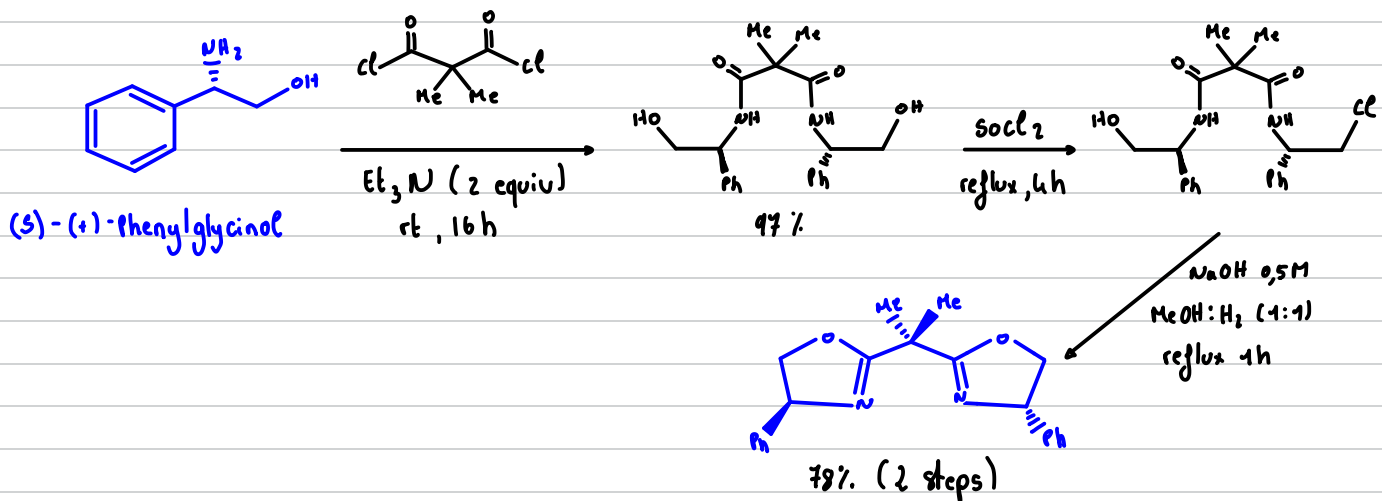


R = H, Me
 R-R = -(CH₂)₂₋₅



R = Ph, t-Bu, i-Pr, i-Bu

1.2. Synthesis of chiral ligands and of bis(oxazoline)-metal complexes

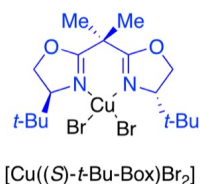
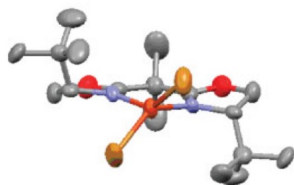


M = Cu(II), Zn(II), Ni(II), Sn(II)
 Ti(V)...

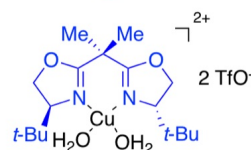
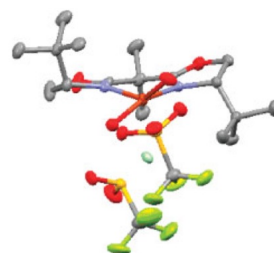
il est importante de déterminer la géométrie du métal dans ces box.

1.3. Structure of bis(oxazoline)-metal complexes

Exemples of box-Cu(II) complexes

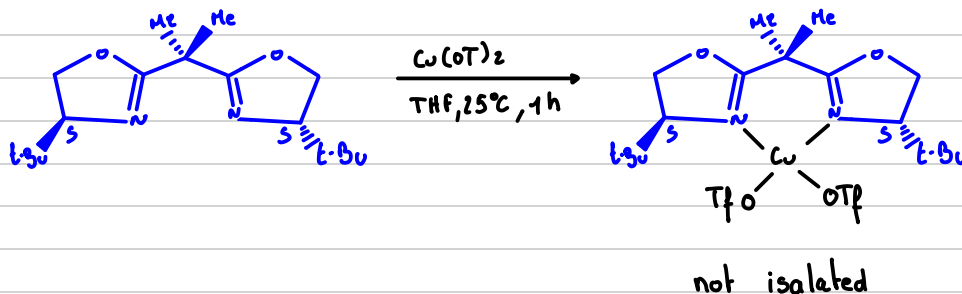


distorted square planar copper geometry

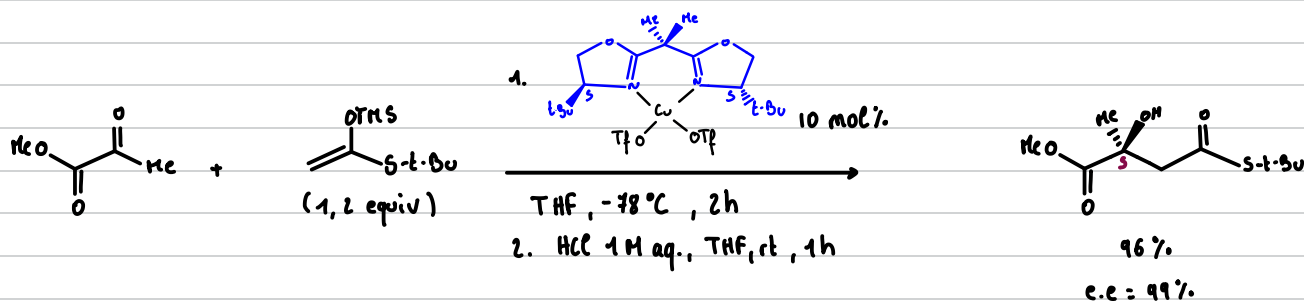


distorted square pyramidal copper geometry

1.4. Catalytic enantioselective Mukaiyama aldol reactions



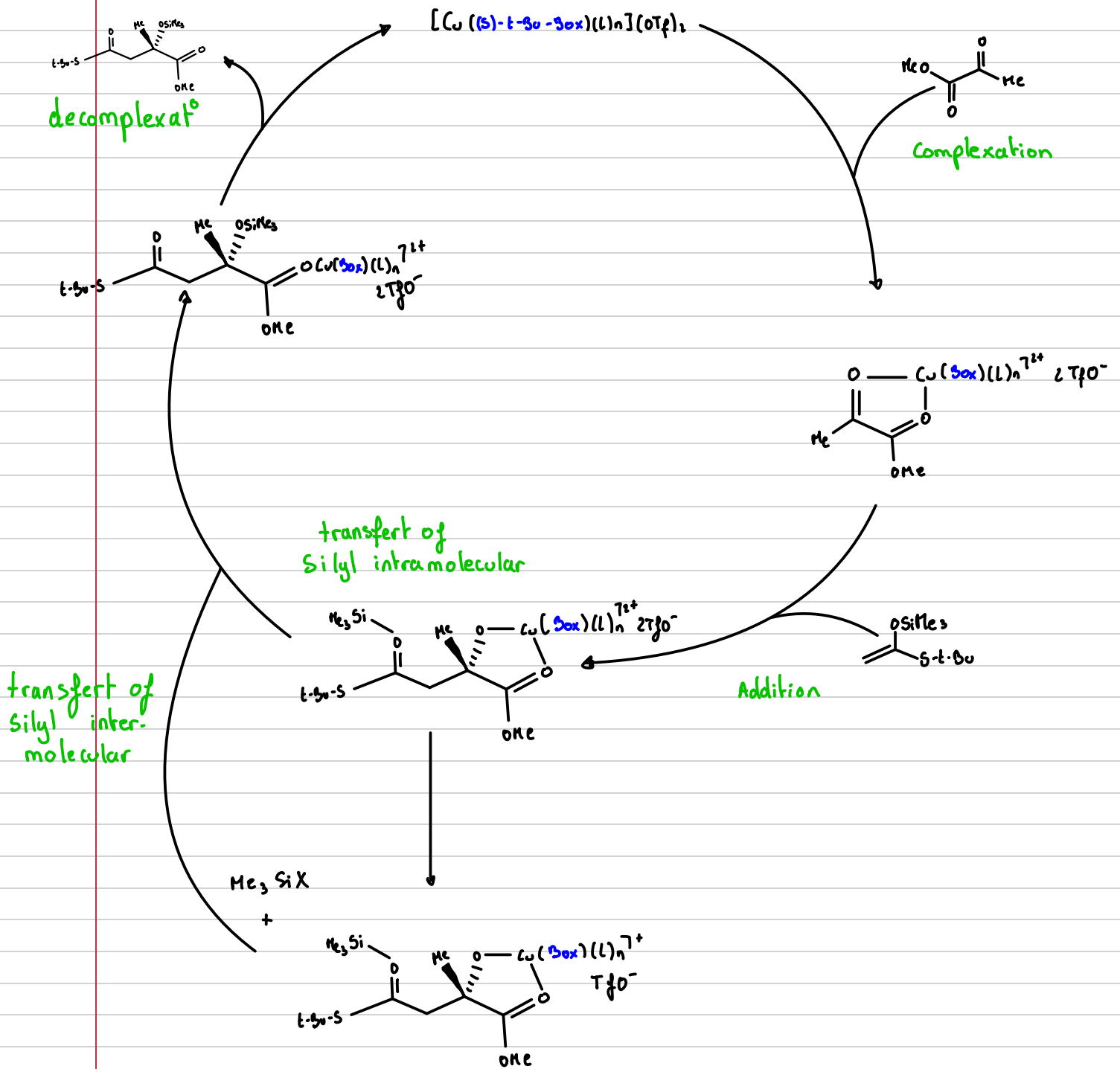
c'est la 1^{re} réact°
de Mukaiyama enantioselective (1999)



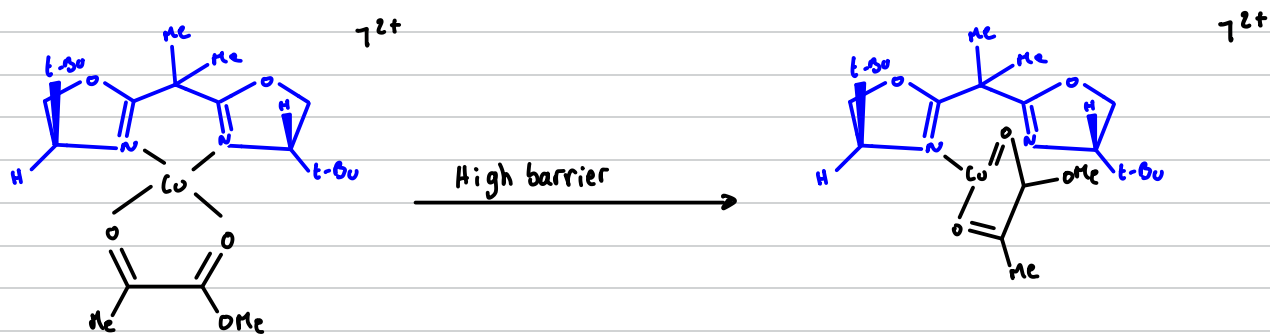
effect of temperature on enantioselectivity

-78 °C	e.e. = 99%
-50 °C	e.e. = 97%
-20 °C	e.e. = 95%
+20 °C	e.e. = 92%

Mechanism

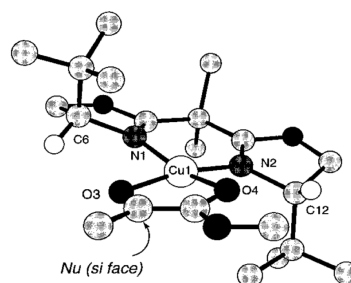
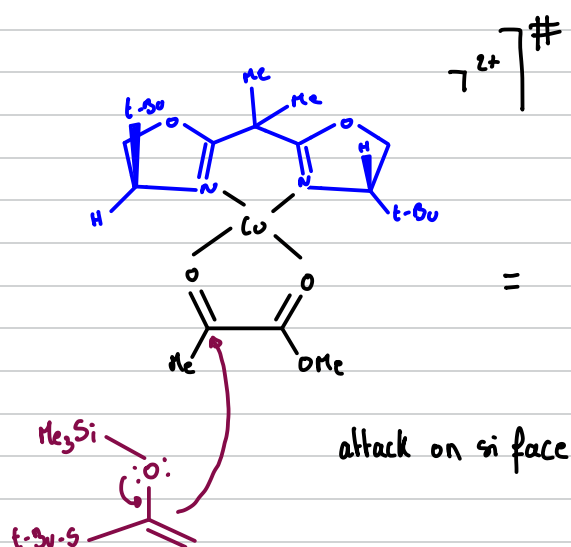


Proposed stereochemical model



square planar Cu(II) geometry

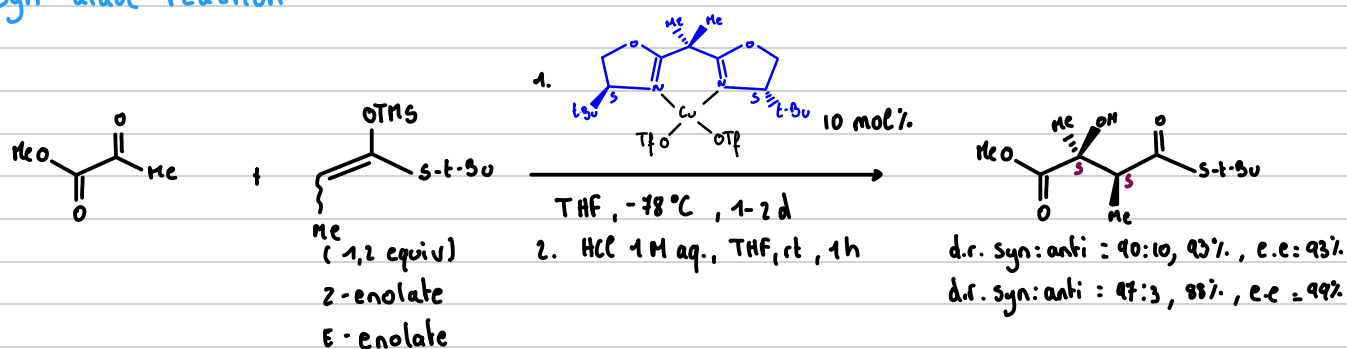
tetrahedral Cu(II) geometry

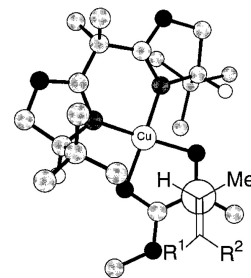
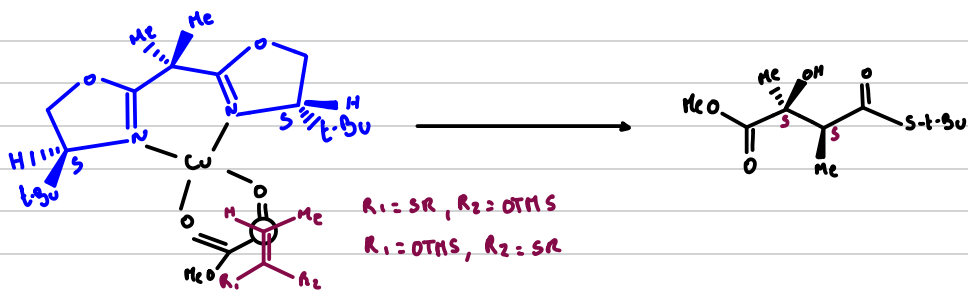


distorted square planar copper geometry

Computational structure of $[\text{Cu}((S)\text{-t-Bu-Box})(\text{pyruvate})]^{2+}$
(PM3 semiempirical calculations)

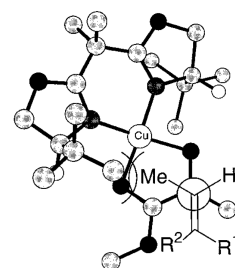
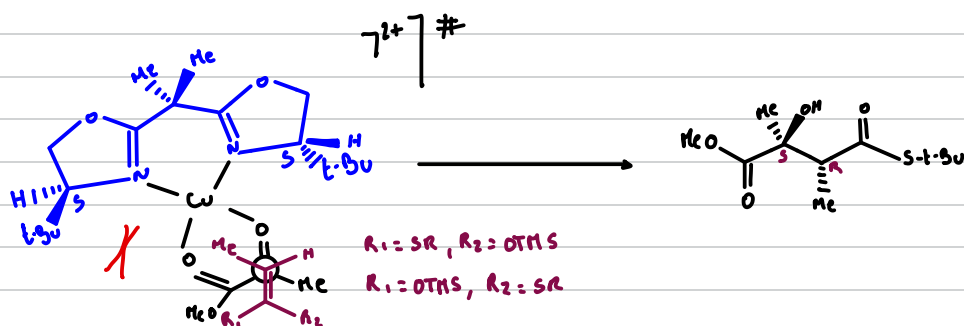
Syn aldol reaction





Attack favored on si face

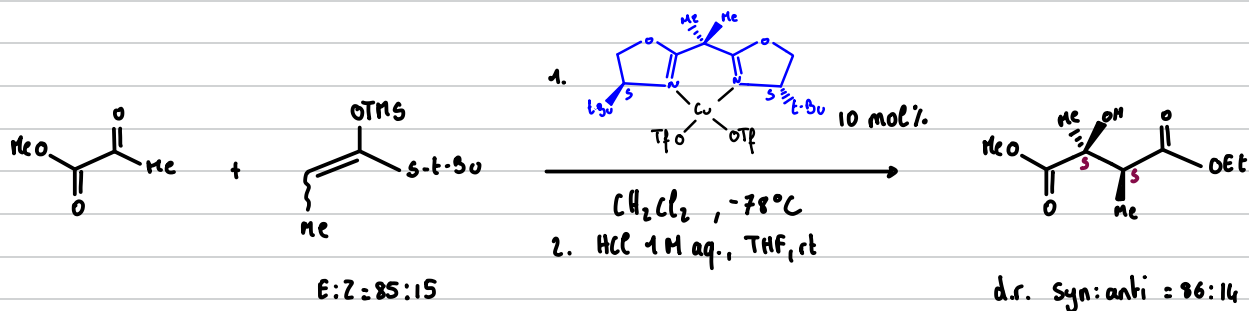
Attack favored on re face

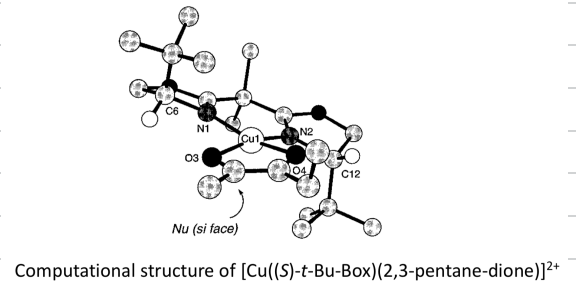
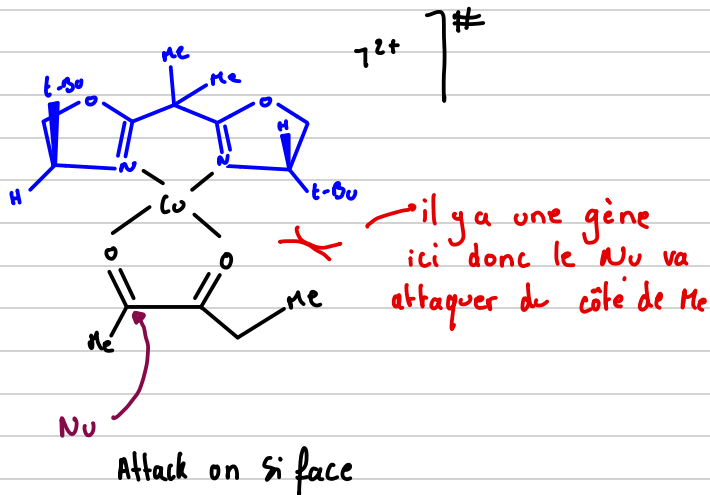
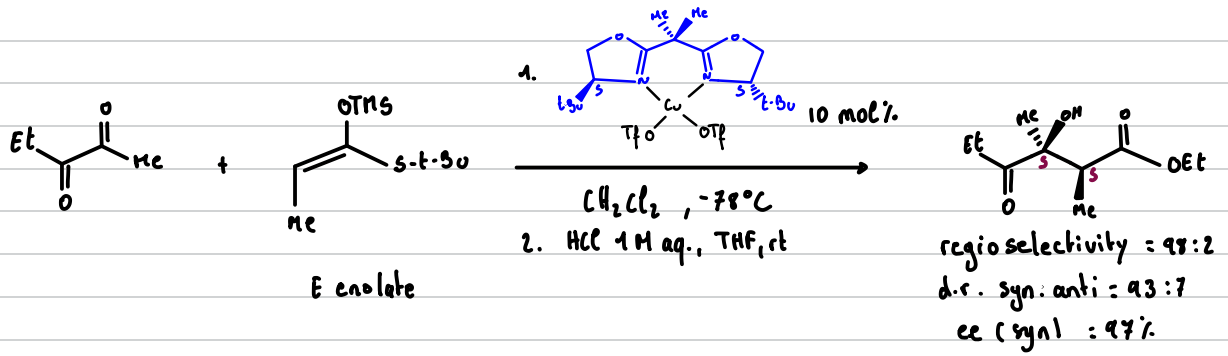


Attack favored on si face

Attack favored on si face

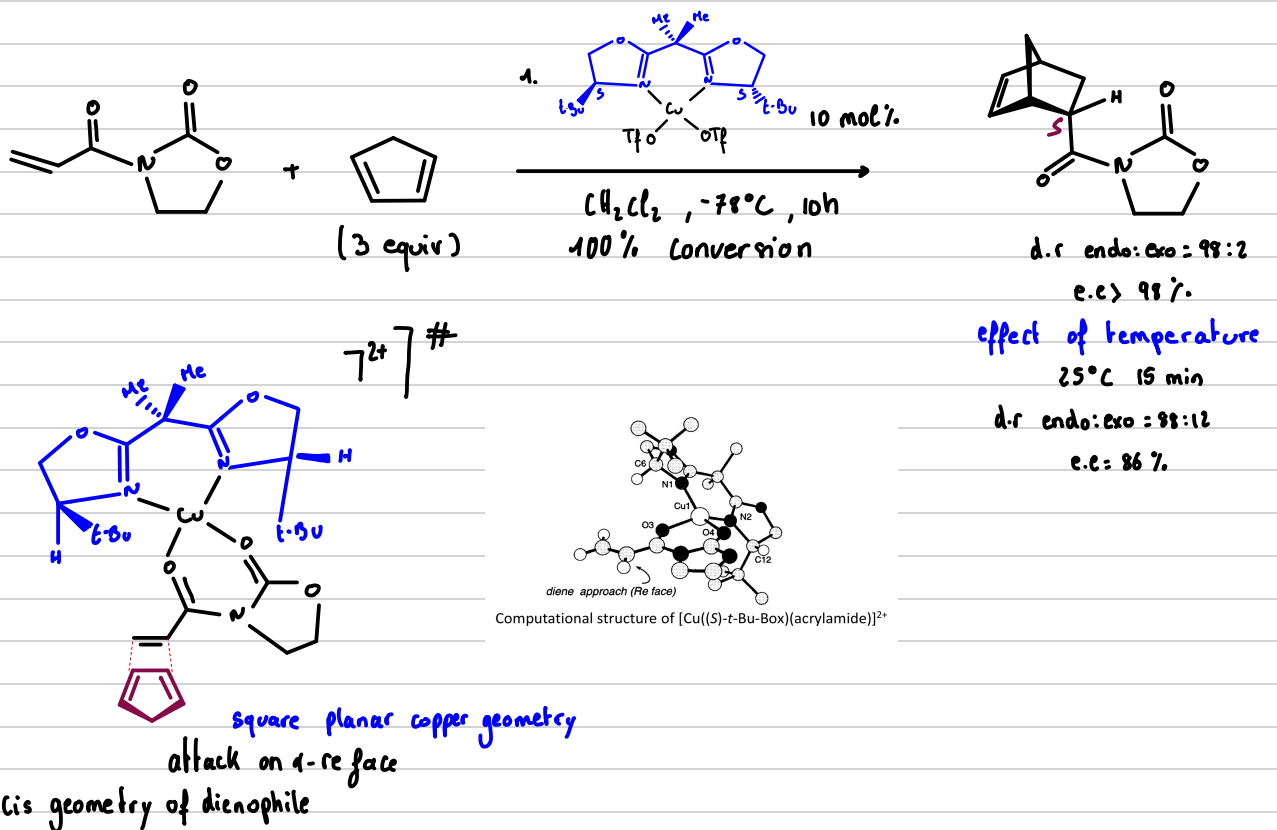
il y a une gêne stérique entre t -Bu devant et le Me



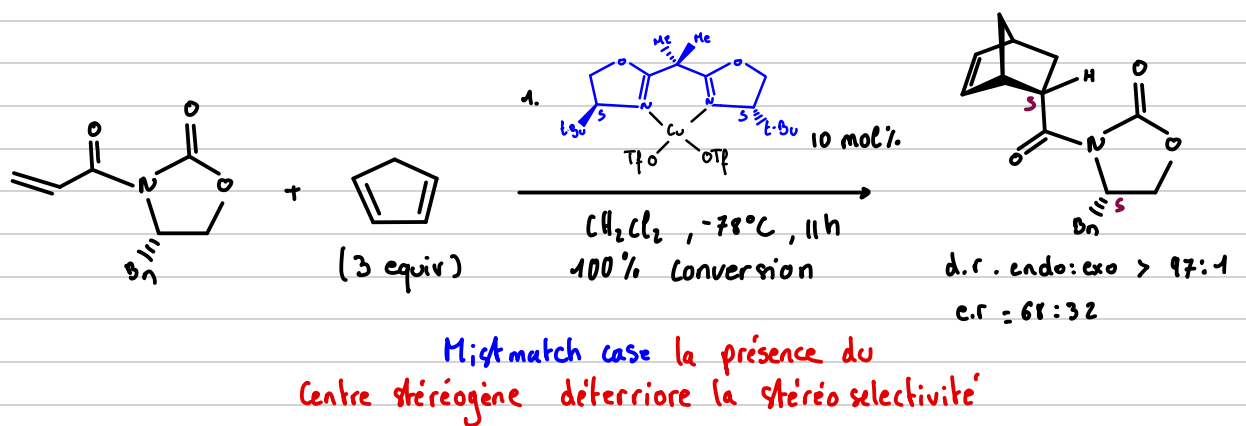
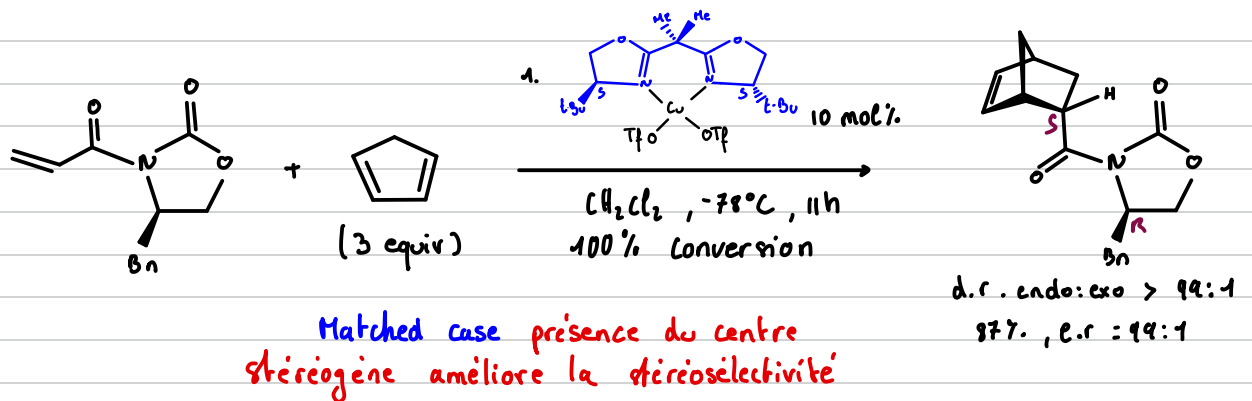


⚠ Reviser la nomenclature Re/Si.

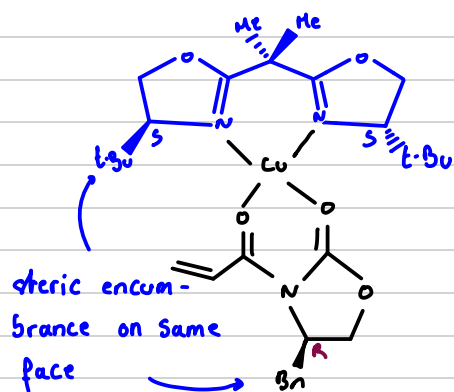
1.5. Catalytic enantioselective Diels-Alder reactions



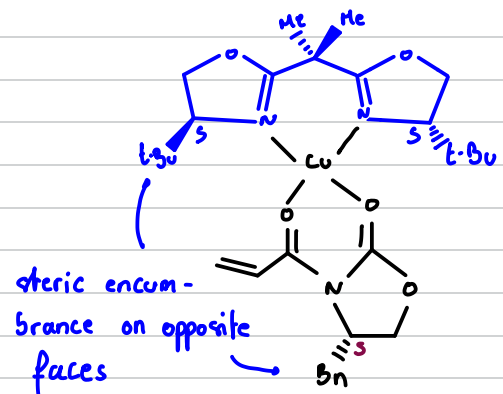
Match/Mismatch effects



explicat°

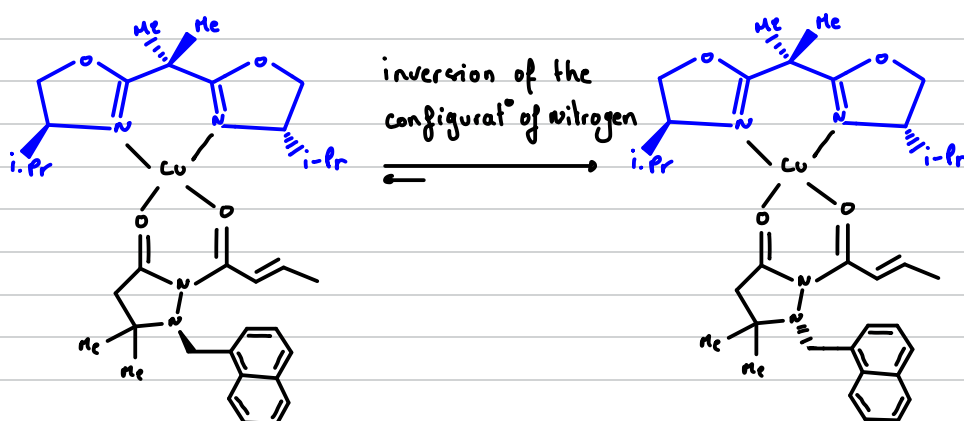
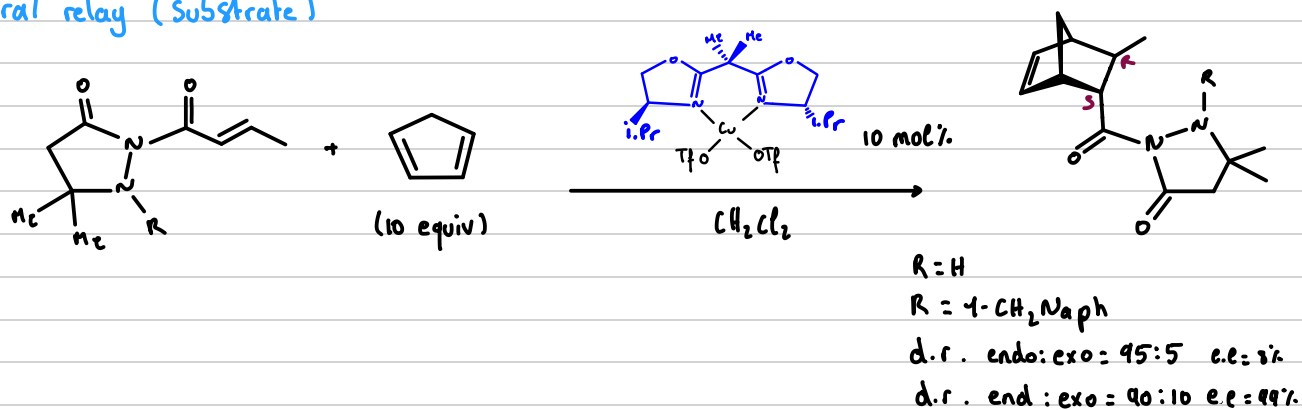


Matched case

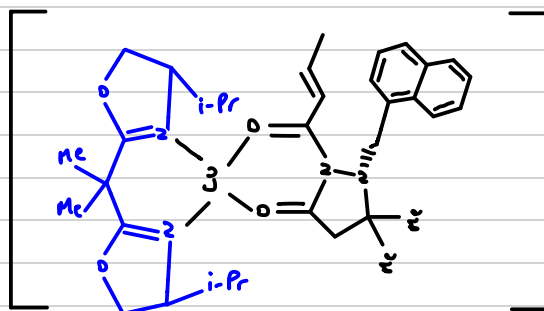


Mismatch case

Chiral relay (substrate)



Achiral template relays and amplifies the stereochemistry from ligand



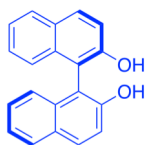
attack favoured from (4)-Re face

References :

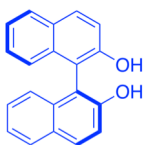
- Desimoni G., Faita G., Jorgensen K.A. *Chem. Rev.* **2011**, 111, PR284-PR437.
- Corey E.J., Imai N., Zhang H.Y. *J. Am. Chem. Soc.* **1991**, 113, 728-729.
- Evans D.A., Woerpel K.A., Nosse B., Schall A., Shinde Y., Jezek E., Haque M.M., Chhor R.B., Reiser O. *Org. Synth.* **2006**, 83, 97-102.
- Evans D.A., Burgey C.S., Kozlowski M.C., Tregay S.W. *J. Am. Chem. Soc.* **1999**, 121, 686-699.
- Evans D.A., Mille, S.J., Letcka T., von Matt P. *J. Am. Chem. Soc.* **1999**, 121, 7559-7573.
- Sibi M.P., Venkatraman L., Liu M., Jasperse C.P. *J. Am. Chem. Soc.* **2001**, 123, 8444-8445.

2. 1,1'-Binaphthyl-2,2'-diols Chiral ligands (BINOL)

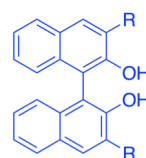
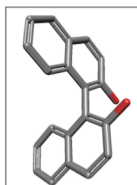
2.1. Structure of 1,1'-Binaphthyl-2,2'-diols chiral ligands



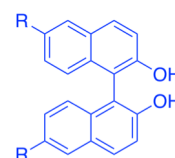
(R)-(+)-BINOL



(S)-(-)-BINOL



3,3'-disubstituted BINOL



6,6'-disubstituted BINOL

Noyori (1979): chiral ligand in stoichiometric amount

References :

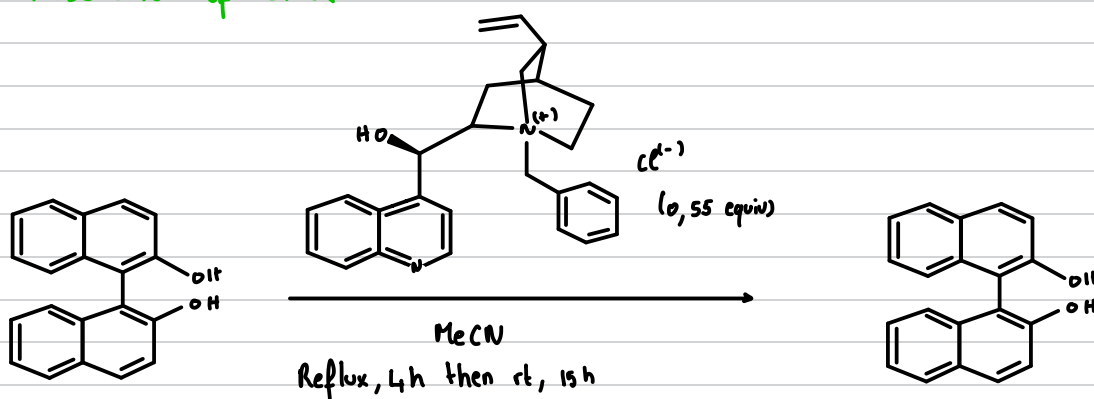
Brunel J.M. Chem. Rev. **2007**, 107, PR1-PR45.

Chen Y., Yekta S., Yudin A.K. Chem. Rev. **2003**, 103, 3155-3211.

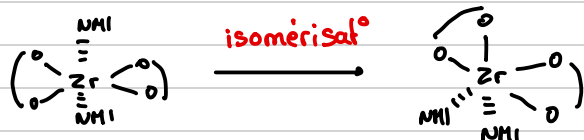
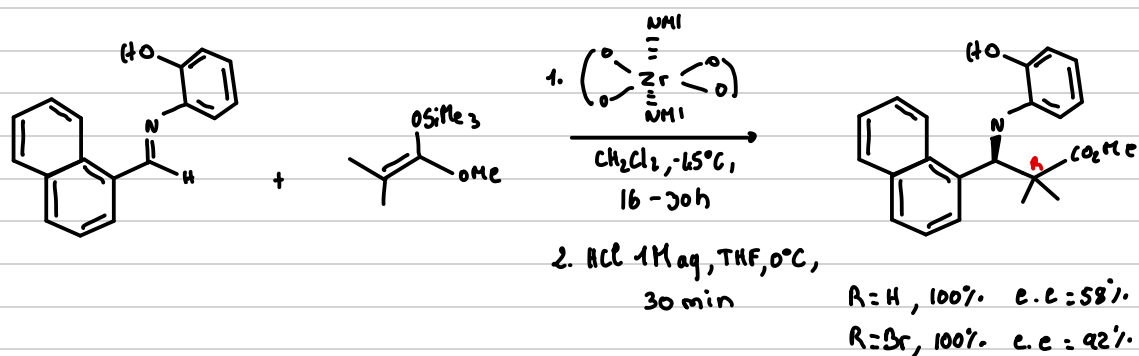
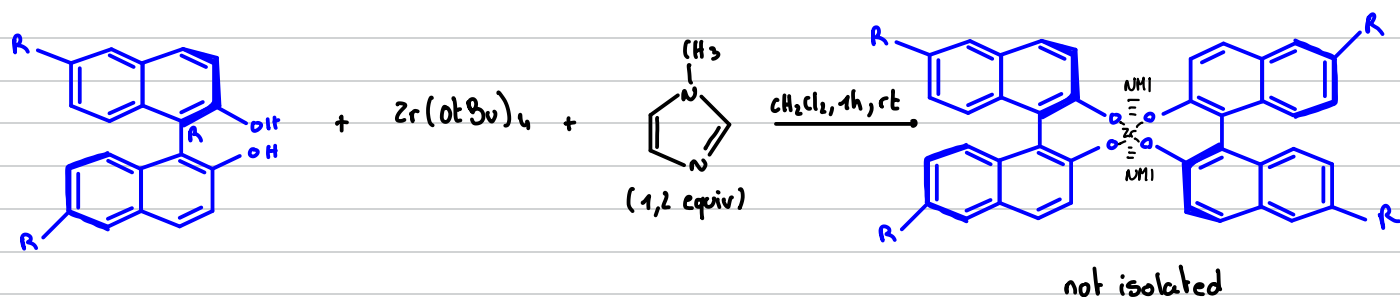
Cai C.D., Hughes D.L., Verhoeven T.R., Reider P.J. Org. Synth. **1999**, 76, 1.

Ishitani H., Ueno M., Kobayashi S. J. Am. Chem. Soc. **2000**, 122, 8180-8186.

2.2 Resolution of Binol

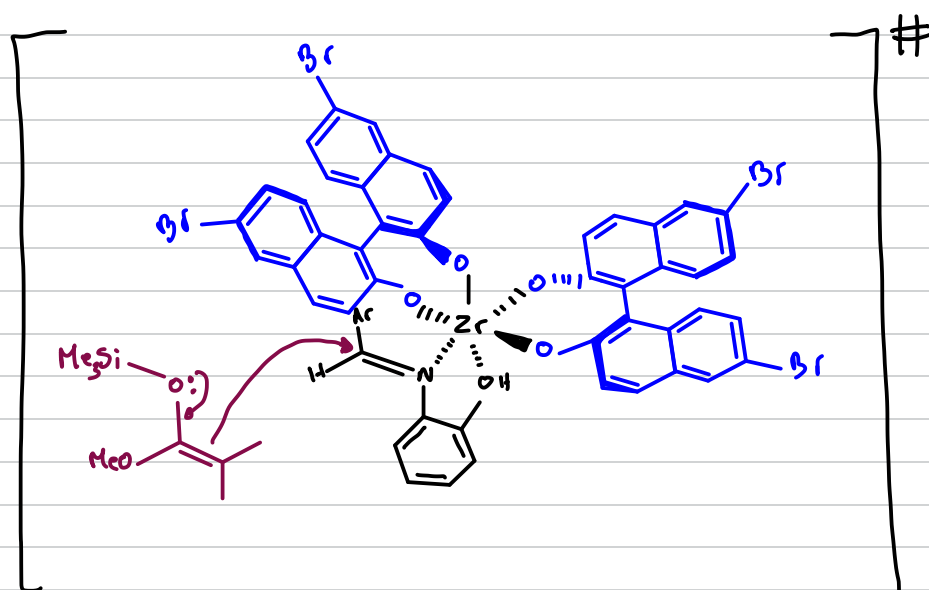


2.3. Aza-aldol reaction (Mannich reaction)



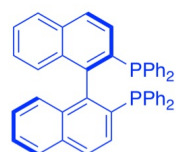
étape 1: isomérisation
il faut que les 2 azotes soient côte à côte.

Proposed stereochemical model



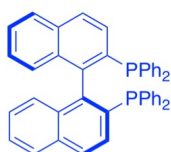
3. 2,2'-diphenylphosphino-1,1'-binaphtyls (BINAP)

3.1. Structure of 2,2'-diphenylphosphino-1,1'-binaphtyls Chiral ligands

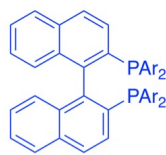


(R)-(+)-BINAP

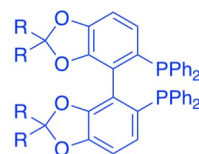
Noyori (1980)



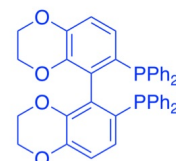
(S)-(-)-BINAP



TolBINAP $\text{Ar} = 4\text{-CH}_3\text{C}_6\text{H}_4$
XylBINAP $\text{Ar} = 3,5\text{-(CH}_3)_2\text{-C}_6\text{H}_3$



SEGPHOS $\text{R} = \text{H}$
Difluorophos $\text{R} = \text{F}$

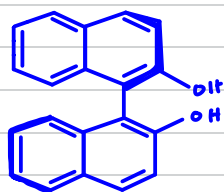


SYNPHOS

axial chirality

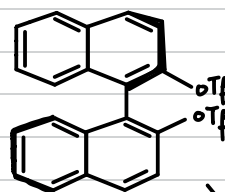
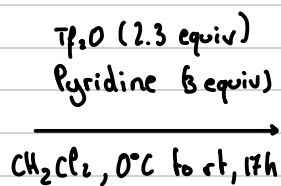
C₂-symmetric ligand

3.2. Synthesis of chiral ligands



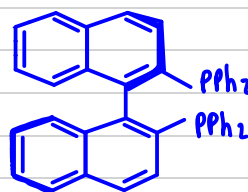
(R)-BINOL

obtained by resolution
of racemic Binol



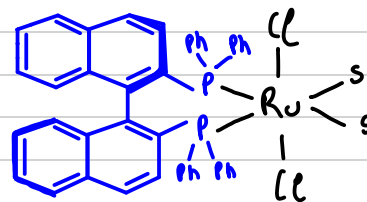
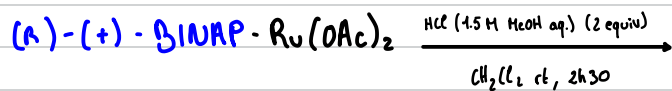
94%

Ph_2PH
 $\text{NiCl}_2\text{ dppf (10 mol\%)}$
 DABCO (4 equiv)
 100°C, 3d

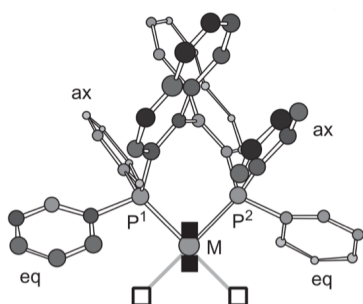


77%
(R)-BINAP

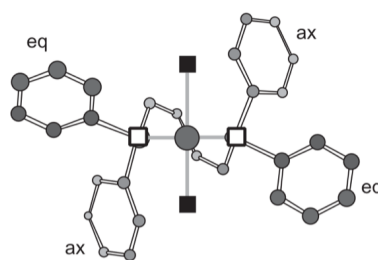
3.3. Asymmetric hydrogenation of ketones



$(R)-(+)-BINAP \cdot RuCl_2$
 (polymeric form)

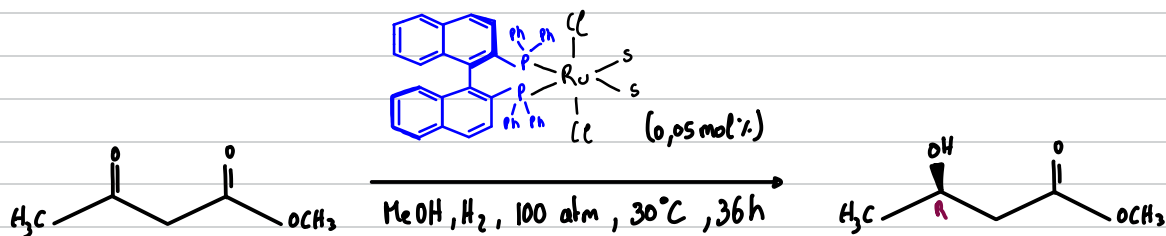
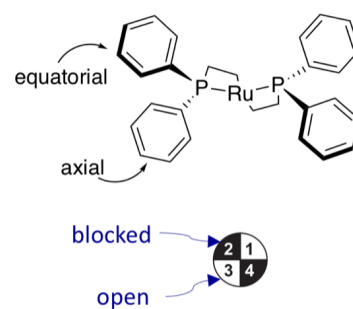


top view

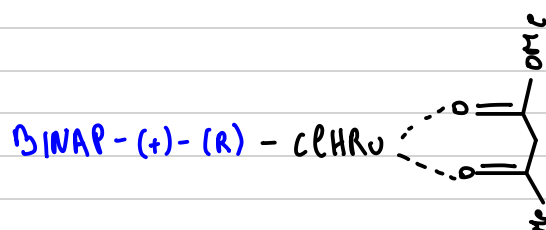


side view

□ = coordination site in the P^1-Ru-P^2 plane
 ■ = coordination site out of the P^1-Ru-P^2 plane

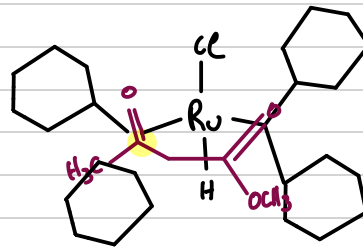
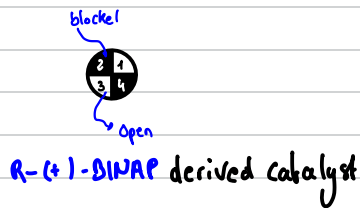


99%, e.e. > 99%.

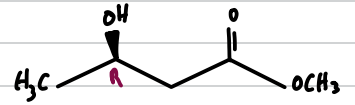


intermédiaire chélaté.

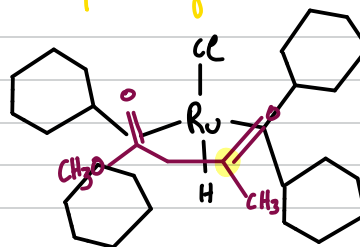
Proposed stereochemical model



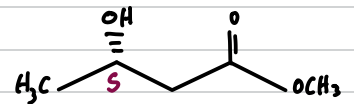
Attack on si face
favoured



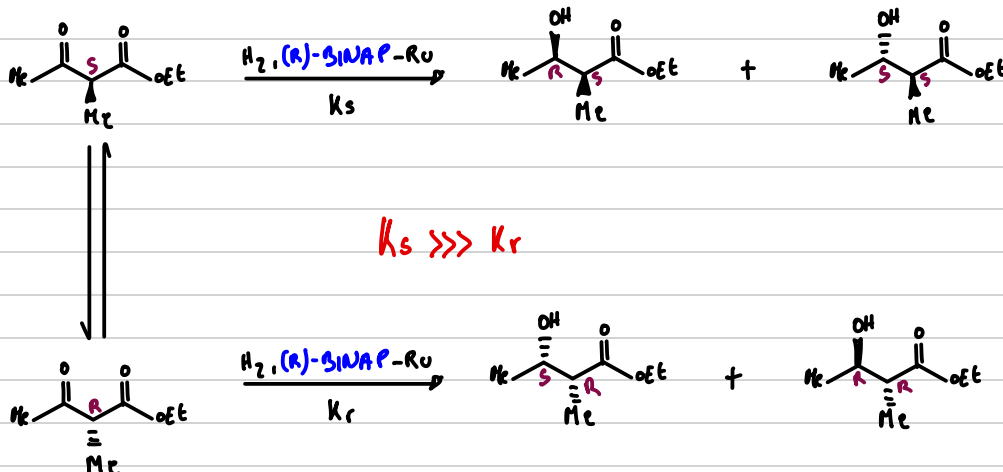
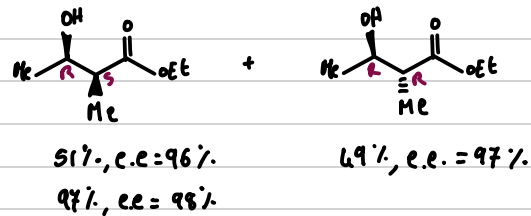
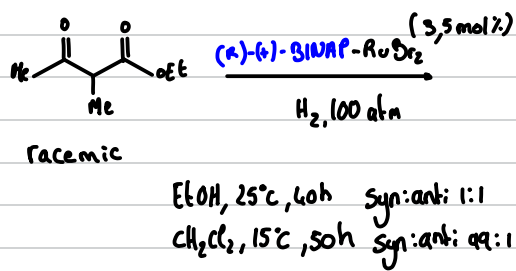
fonct° cétone
qui réagit



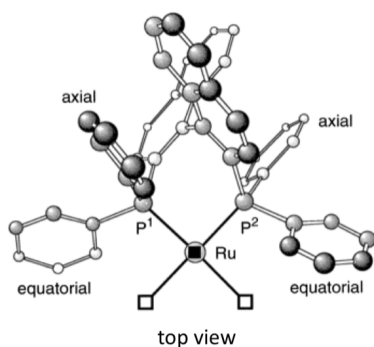
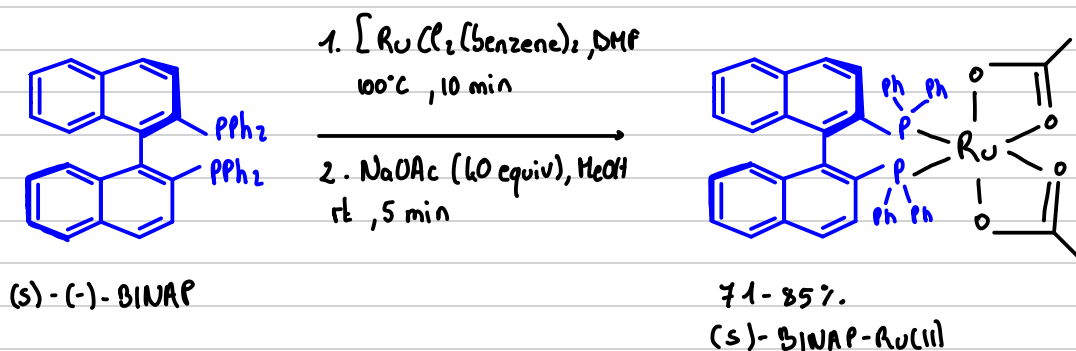
unfavoured



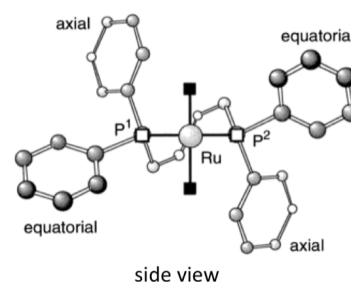
Dynamic Kinetic Resolution (DKR)



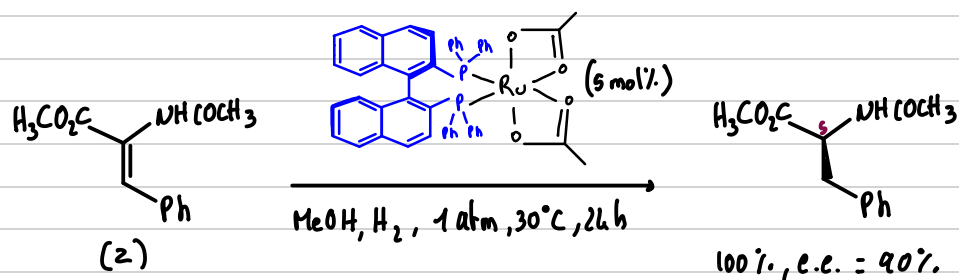
3.4. Asymmetric hydrogenation of enamides



□ = coordination site in the P^1-Ru-P^2 plane
 ■ = coordination site out of the P^1-Ru-P^2 plane



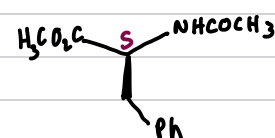
2 1
 3 4



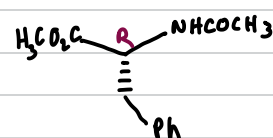
Proposed stereochemical mode



attack on Re face
 favoured



attack on Si face
 unfavoured



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- Noyori R., Ohkuma T., Kitamura M., Takaya H., Sayo N., Kumabayashi H., Akutagawa S. *J. Am. Chem. Soc.* **1987**, 109, 5856-5858.
- Noyori R., Ohkuma T., Tokunaga M., Kitamura, M. *Bull. Chem. Soc. Jpn.* **1995**, 68, 36-56.
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